MODEL DELIVERABLE

VC Scout

Model Definition and Initial Results

Startup Growth and Investment | Unicorn Companies Dataset

1,060	7	80 / 20	5-fold	0.51
Companies modeled	Models compared	Train / test split	Cross validation	Held-out test R^2

Team 17 Mia Murphy • Finn Kliewer • Kayvon Jafarzadeh • Nathanael Gill • Om Patel

Goal: define the modeling toolkit for the deeper dive, show a working implementation on the cleaned unicorn dataset, and compare candidate models before locking the champion for the VC Scout framework.

MODELING OBJECTIVE

What we model, and which tools we use

The deeper dive moves from description to supervised prediction, framed strictly as conditional-on-unicorn pattern learning.

The prediction task

Target: ln(Valuation in $B) for each unicorn company.

- Regression, not classification: every row is already a unicorn, so "success vs failure" cannot be learned from this data. We instead predict the valuation scale a unicorn reaches.
- The log transform tames the long right tail found in EDA (median $2B, max $180B) so errors are measured in relative terms.
- Model outputs feed VC Scout: predicted valuation given funding, sector, and geography becomes the benchmark, and companies or segments above the benchmark flag efficient growth signals.
- Framing stays non-causal: coefficients and importances are read as associations among historical unicorns, never as guarantees.

Optimization, simulation, and other tools

Supervised ML (scikit-learn)

Core tool. Regularized linear models and tree ensembles inside leakage-safe Pipeline objects.

Grid-search optimization

Hyperparameters chosen by exhaustive GridSearchCV over 5-fold cross validation. This is the optimization component of the project.

Bootstrap simulation

2,000 resamples of the held-out test set simulate the sampling distribution of model accuracy and give a confidence interval.

Weighted scoring framework

Model residuals and importances convert into transparent VC Scout sector and region scores in the next phase.

DATA & FEATURES

Variable definitions, feature engineering, and sampling

Every engineered field is auditable, and every excluded row is counted, consistent with the EDA cleaning rules.

Variable	Role	Definition and engineering
ln(Valuation)	Target	Natural log of valuation parsed from strings like "$180B" into numeric $B.
ln(Funding)	Feature	Natural log of disclosed funding; $M values converted to $B before the log.
Industry	Feature	15 standardized sectors; the AI capitalization duplicate is merged.
Continent	Feature	Headquarters region, 6 categories, one-hot encoded.
Era	Feature	Market regime cohort: Pre-2021, 2021, Post-2021, from EDA finding H1.
Years to unicorn	Feature	Unicorn year minus founding year; speed of the growth path.
Investor count	Feature	Count of names listed in the Select Investors field.

Preprocessing and sampling

- One-hot encoding for categoricals; standardization for linear and KNN models only.
- Sampling: 80/20 train/test split (848/212, seed 17); 5-fold CV inside the training set.
- All transforms fit on training folds only, inside sklearn Pipelines, to prevent leakage.

Exclusions: 1,074 to 1,060 rows

- 12 rows with unknown funding and 1 with $0 funding (log undefined).
- 1 negative years-to-unicorn date anomaly flagged in EDA.
- Funding efficiency is NOT a feature: it contains the target (valuation / funding), so using it would leak the answer.

MODEL DEFINITIONS

Why this set of models: interpretability plus flexibility

Each candidate answers a different question about the data; together they bracket the bias-variance tradeoff.

Baseline (mean) *Sanity floor*

Predicts the training mean. Any real model must beat $R^2 = 0$.

Linear OLS *Interpretability*

The log-log regression planned in EDA. Its funding coefficient is an elasticity: the % change in valuation per 1% more funding.

Ridge & Lasso *Stability*

Regularization handles 20+ correlated one-hot columns; Lasso also zeroes weak features, acting as automatic feature selection.

KNN *Local structure*

Nonparametric check: predicts from the most similar unicorns, with no functional form assumed.

Random Forest *Nonlinearity*

Bagged trees capture sector-by-funding interactions and thresholds that linear models miss.

Gradient Boosting *Champion candidate*

Sequential shallow trees, strongest expected fit on tabular data of this size; still explainable via permutation importance.

Selection logic

EDA showed a skewed target, strong sector and geography segmentation, and a suspected nonlinear funding relationship. That points to tree ensembles for accuracy, while the linear family is kept as the interpretability companion whose coefficients VC Scout can explain to users. KNN and the baseline guard against overclaiming: if a simple method matched the ensembles, the extra complexity would not be justified.

IMPLEMENTATION

How the models are built: split, tune, validate

One leakage-safe pipeline per model; every hyperparameter is chosen by cross validation, never by peeking at the test set.

Clean + engineer
1,074 raw rows to 1,060 modeled

→

Split 80/20
848 train / 212 held-out test

→

Pipeline prep
encode + scale inside CV folds

→

GridSearchCV
5-fold CV over 99 combinations

→

Final test
one look at the 212 test rows

Model	Hyperparameter grid searched	Best setting (5-fold CV)
Ridge	alpha in {0.1, 1, 3, 10, 30, 100}	alpha = 30
Lasso	alpha in {0.0005, 0.001, 0.005, 0.01, 0.05}	alpha = 0.005
KNN	k in {5, 10, 15, 25, 40}; uniform or distance weights	k = 25, uniform
Random Forest	300/600 trees; depth {4, 6, 10, none}; leaf {2, 5, 10}	600 trees, depth 4, leaf 2
Gradient Boosting	200/400/800 trees; rate {0.01, 0.03, 0.1}; depth {2, 3, 4}; subsample {0.8, 1.0}	200 trees, rate 0.03, depth 2, subsample 0.8

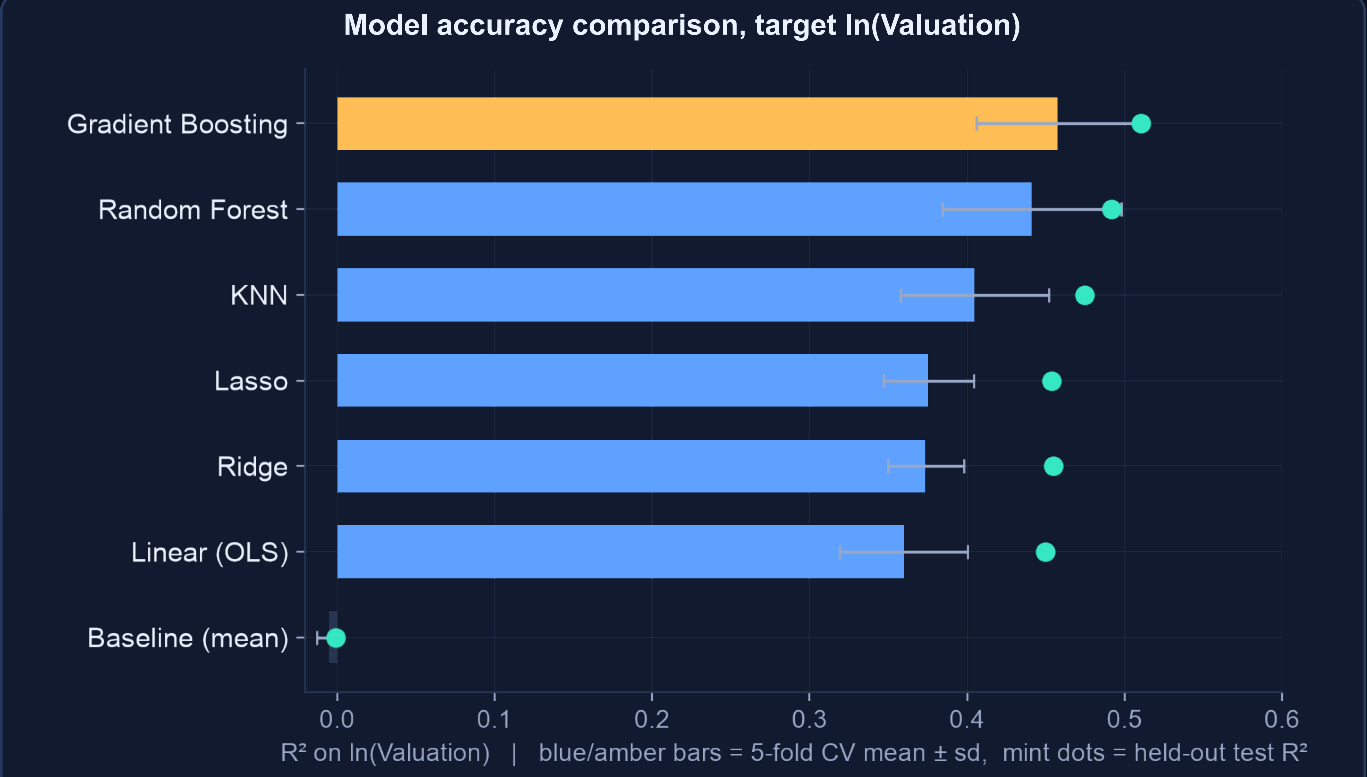
Validation discipline

- Same folds (seed 17) for every model, so comparisons are apples to apples.
- Scoring metric: R^2 on ln(Valuation); MAE and RMSE reported alongside.
- The 212 test companies are touched exactly once, after all tuning is frozen.
- The tuned winner prefers shallow, slow-learning trees: evidence that regularization matters more than raw capacity here.

ALTERNATE MODELS

Gradient boosting wins, and the margin is honest

Every alternate model was fit, tuned, and scored on identical folds and the identical held-out test set.



Model	CV R^2	Test R^2	MAE (ln)
Baseline (mean)	-0.01	0.00	0.62
Linear (OLS)	0.36	0.45	0.45
Ridge	0.37	0.45	0.45
Lasso	0.38	0.45	0.45
KNN	0.41	0.47	0.46
Random Forest	0.44	0.49	0.44
Gradient Boosting	**0.46**	**0.51**	**0.44**

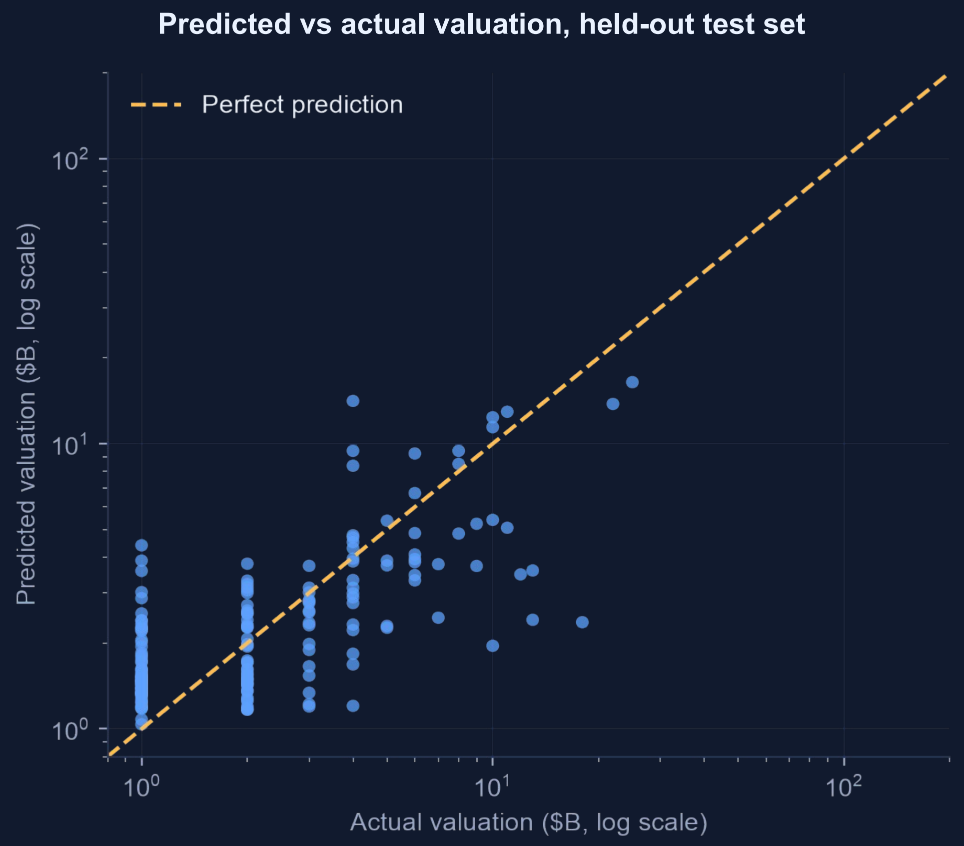
Why the champion is better

- Best CV R^2 (0.46) and best test R^2 (0.51) with the lowest error; the gain over the best linear model (+0.08 CV R^2) comes from sector-funding interactions.
- Beats Random Forest with a 3x smaller, shallower ensemble (200 trees, depth 2 vs 600, depth 4), and its CV-to-test gap shows no overfitting.
- OLS is retained as the explanation layer for VC Scout narratives.

INITIAL RESULTS

The champion predicts valuation scale, not exact price tags

Held-out predictions track actual valuations across two orders of magnitude on the 212 unseen companies.



How to read this

- Points hug the diagonal through the $1B to $10B core where most unicorns live.
- The model compresses mega-outliers: it under-predicts the $100B+ names, which is expected when features stop at funding, sector, geography, and timing.
- Median absolute error is 37% of valuation, versus 77% for the naive baseline: roughly half the error, from public fields alone.

0.51

Test R^2 on ln(Valuation)

0.44

Test MAE in log units

37%

Median abs. % error

77%

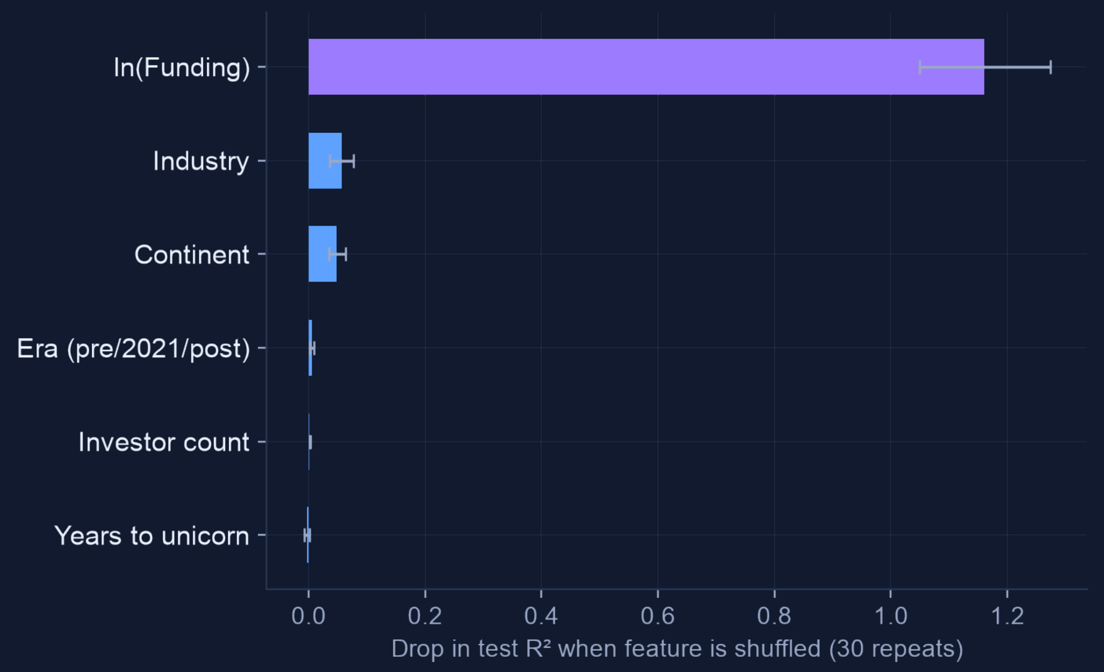
Baseline median error

INITIAL RESULTS

Funding dominates; sector and geography refine the estimate

Permutation importance on the test set shows what the champion actually uses, and the funding curve confirms diminishing returns.

Permutation importance, champion model



Shuffling ln(Funding) destroys 1.16 points of R^2 (the drop exceeds 1.0 because shuffled predictions score below zero). Industry (0.06) and Continent (0.05) add real but secondary signal; era, investor count, and speed add little once funding is known.

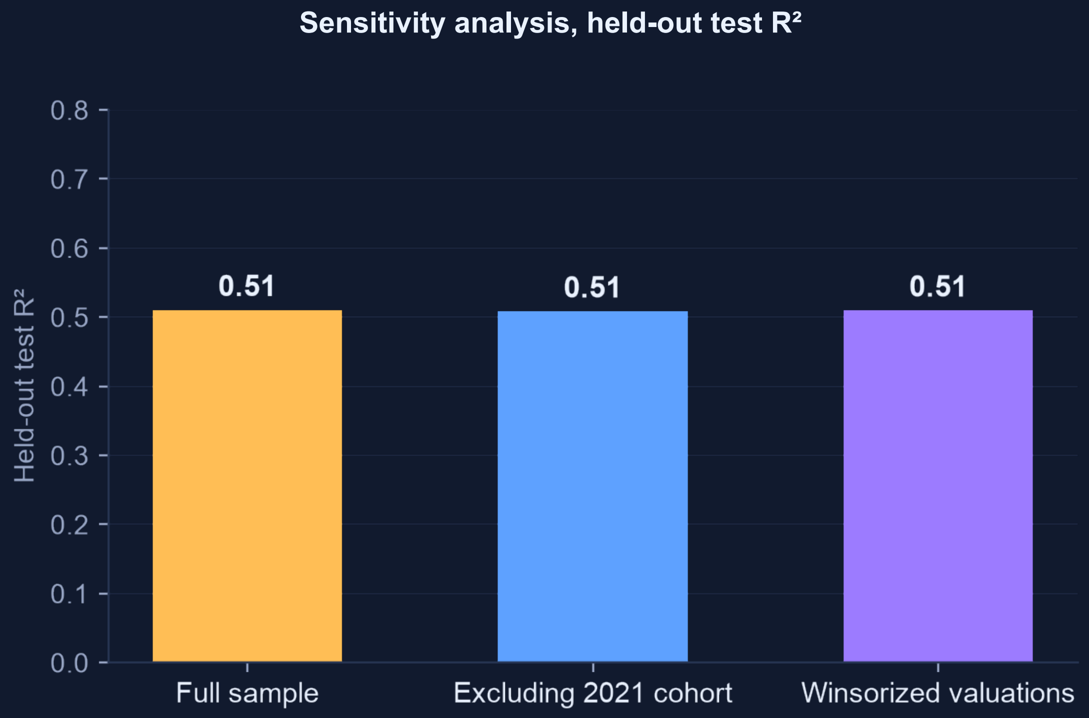
Partial dependence: valuation vs funding

Elasticity ≈ 0.57. From the OLS companion model: a 1% increase in funding is associated with about a 0.57% higher valuation. Below 1.0 means diminishing returns, confirming EDA hypothesis H3: capital scales valuation, but sub-linearly.

ROBUSTNESS

The result survives the 2021 regime and the mega-outliers

The two biggest threats flagged in EDA, the 2021 cohort and the long tail, barely move the champion's accuracy.



Retraining without the 520-company 2021 cohort, or with winsorized valuations, changes test R^2 by less than 0.01.

Bootstrap simulation of accuracy

- 2,000 bootstrap resamples of the 212 test companies simulate how accuracy varies across samples.
- 95% confidence interval for test R^2: 0.39 to 0.61.
- Even the pessimistic bound beats every linear model's cross-validated R^2, so the champion choice is not a lucky split.

What this settles from the EDA plan

- H1 (2021 is a distinct regime): true for volume, but it does not distort the valuation model.
- H3 (log-scale funding effect with diminishing returns): confirmed, elasticity about 0.57.
- Robustness checks promised in the EDA deck are now implemented, not just planned.

NEXT STEPS

From a tuned model to the VC Scout framework

The champion model becomes the engine that scores sectors and regions; the limitations stay on the label.

1 Benchmark engine

- Champion predictions define the expected valuation for any funding-sector-geography profile.
- Segments that consistently beat their benchmark (positive residuals) become high-signal VC Scout categories.

2 Scoring layer

- Blend model-based signals with EDA descriptives: counts, median efficiency, and time-to-unicorn.
- Weights stay transparent and user-adjustable, per the charter's explainability requirement.

3 Hardening

- Add investor network features (top-fund co-mention flags) and test whether they lift R^2.
- Quantile gradient boosting for prediction intervals, so VC Scout can show ranges instead of false precision.

Limitations we will keep saying out loud

This model explains valuation variation among companies that already became unicorns. It cannot rate an arbitrary startup's probability of success, because failed and non-unicorn startups are absent (survivorship bias). All findings are associations conditional on unicorn status, and VC Scout will present them as scouting signals with confidence ranges, not as predictions or investment advice.